B.Tech. I Sem : EE, ME, EI, EC, IoT
Applied Physics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Write down the postulates of planck's quantum hypothesis. 02

(b) Define the phase velocity and group velocity. 02

(c) What is wave function, write its properties. 03

OR

A microscope using photon is employed to locate an electron in an atom to within a distance 2*10 (-11) meter. What is the un certainity in the momentum of electron located in this way? 03

(d) Deduce an expression for the compton shift. 05

OR

Obtain scherodinger time dependent wave equation. 05

Q.2 (a) What are the main properties of LASER? 02

(b) Write the two important applications of LASER. 02

(c) Calculate the numerical aperture and acceptance angle for an optical fiber, given that refractive index of the core and the cladding is 1.41 and 1.40 respectively. 03

OR

Explain population inversion process. 03

(d) Discuss the He – Ne LASER on the basis of ray diagram, construction and working. 05

OR

Explain the construction and re construction of image on hologram. 05

Q.3 (a) What is effective mass? 02

(b) Define mobility of charge carriers. 02

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(c) Explain the properties of PN junction. 03

OR

Differentiate between direct and indirect band gap of semiconductor. 03

(d) Explain the radiative and non radiative recombination mechanism in semiconductors. 05

OR

Explain the structure and characteristics of LED. 05

Q.4 (a) Define superconductor. 02

(b) Differentiate type I and Type II superconductors. 02

(c) What are the applications of superconductors? 03

OR

What are the applications of nanotechnology? 03

(d) Discuss the BCS theory of superconductors. 05

OR

Explain the properties and uses of fullerene. 05

Q.5 (a) Differentiate between polar and non polar dielectric. 02

(b) Define dielectric constant 02

(c) What is piezoelectric effect? Give two application of piezoelectric material. 03

OR

What is piezo ceramics? Name the two mostly used piezo ceramics material. 03

(d) Obtain the expression for gauss's law in dielectric. 05

OR

Define ferroelectric material. Write its properties. 05

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